Interpretable Concept Bottleneck Models for Reliable Diabetic Retinopathy Grading

***Abstract*—Diabetic retinopathy (DR) is a leading cause of preventable blindness, necessitating reliable and interpretable automated screening systems. While conventional deep learning models achieve high classification accuracy, they predominantly function as opaque black boxes, limiting clinical trust and human-in-the-loop verification. This paper presents an Interpretable Concept Bottleneck Model (CBM) for DR grading. The proposed two-stage architecture maps retinal images to an intermediate, human-readable layer of four clinical concepts: Microaneurysms (MA), Hemorrhages (HE), Hard Exudates (EX), and Soft Exudates (SE), before computing a final diagnostic grade. An EfficientNet-B0 backbone fine-tuned on the IDRiD dataset extracts concept probabilities, which are passed to a transparent linear classifier mapping concepts to four severity grades. Evaluated on the DDR test set (N=4,105), the model achieves a baseline accuracy of 64.21% and a Quadratic Weighted Kappa (QWK) of 0.6867. Crucially, every prediction is explicitly linked to identifiable lesion manifestations, providing a structurally auditable decision chain, a vital step toward accountable medical AI.**

***Index Terms*—Diabetic Retinopathy, Concept Bottleneck Model, Interpretable AI, Clinical AI, DDR Dataset, IDRiD Dataset, EfficientNet-B0, XAI.**

I. Introduction

Diabetic retinopathy (DR) is a predominant cause of preventable visual impairment globally [1]. The growing burden of diabetes necessitates scalable, trustworthy automated screening systems to support ophthalmologists in early grading and intervention. Clinical DR progression is characterised by identifiable retinal lesions: Microaneurysms, Hemorrhages, Hard Exudates, and Soft Exudates, precisely defined in the international clinical grading scale [2]. This tight coupling between visible lesion presence and diagnostic grade makes DR uniquely suited for concept-driven, interpretable artificial intelligence.

Despite significant accuracy gains from deep learning [3], the dominant paradigm remains end-to-end black-box classification: a model maps a retinal image directly to a grade without exposing any intermediate reasoning. This opacity is clinically problematic. Ophthalmologists cannot verify, audit, or override a prediction they cannot inspect. Transfer learning and ensemble methods have improved grading performance [4] but have not resolved this fundamental transparency deficit. Post-hoc tools such as Grad-CAM [5] provide visual hints after the fact, yet they cannot enable a clinician to intervene in the model's reasoning.

The principled solution is the Concept Bottleneck Model (CBM) [6]. Unlike black-box approaches, a CBM forces the network to route every prediction through an intermediate layer of human-defined clinical concepts before producing a final grade. In the DR context, the model must first estimate the presence of MA, HE, EX, and SE, exactly the lesions used in manual grading, and only then assign a severity grade. The result is a fully auditable, correctable, and clinically aligned decision chain. To the best of our finding, this architecture is the most appropriate for reliable DR grading and demonstrates it for the first time in a cross-dataset setting.

II. Literature Review

A. Black-Box Deep Learning for DR

Early automated DR screening relied on hand-crafted features and traditional image processing. The landmark study by Gulshan *et. al.* demonstrated that CNNs trained on large fundus datasets can match ophthalmologist-level accuracy for detecting referable DR [3]. EfficientNet [7] further improved efficiency through compound scaling, and Chilukoti *et. al.* built on this by proposing an ensemble transfer learning framework adopting QWK as the primary ordinal evaluation metric, achieving competitive grading on the APTOS dataset [4]. Anand *et. al.* benchmarked a fine-tuned EfficientNetB0 against several baselines, demonstrating strong adaptability of compact architectures to DR grading tasks [8].

More recently, attention-based and multi-scale methods have pushed accuracy further. Wang *et. al.* proposed a multi-attention residual refinement architecture achieving consistent 2–5% performance gains across multiple CNN families [9], and Zhou *et. al.* demonstrated that self-paced progressive multi-scale training enhances pathological feature discrimination in DR multi-classification [10]. For cross-domain settings, Chen *et. al.* proposed cross-modality transfer learning with an adversarial lesion map generator that improves generalisation across imaging modalities [11], while Ding *et. al.* and Singh *et. al.* further validated robust grading across heterogeneous fundus datasets [12], [13].

Despite these gains, a critical limitation pervades all such systems: none exposes its clinical reasoning. The diagnostic grade is produced without any intermediate representation that a clinician can examine, verify, or correct.

B. Post-Hoc Explainability: Necessary but Insufficient

The XAI community responded to black-box opacity with post-hoc explanation methods. Grad-CAM [5] generates gradient-based saliency maps highlighting influential image regions. While useful for visualisation, these methods do not constrain what the network actually learns. A heatmap may highlight a lesion region coincidentally rather than causally.

Djoumessi *et. al.* demonstrated that an inherently interpretable model explicitly modelling local lesion evidence outperforms post-hoc explanation of a standard DNN in clinical DR screening [14]. Storås and Sundgaard further compared Concept Activation Vectors and CBMs for DR grading, finding CBMs provide more clinically actionable, per-lesion explanations [15]. These findings confirm that post-hoc methods explain model output without constraining the learning process, a fundamental limitation that only architecture-level interpretability can overcome.

C. Why CBM is the Most Reliable Architecture for DR

The CBM framework [6] was designed precisely for settings where the output is determined by a known set of intermediate attributes and human oversight is required. DR grading satisfies both conditions exactly. Table I summarises the key differences between interpretability approaches.

TABLE I
COMPARISON OF INTERPRETABILITY APPROACHES FOR DR GRADING

Property	Black-Box	Grad-CAM	Post-hoc XAI	CBM (Ours)
Clinical concept alignment	×	Partial	Partial	✓
Clinician intervention	×	×	×	✓
Causal explainability	×	×	×	✓
Regulatory compliance	×	Partial	Partial	✓
Cross-dataset transfer	Limited	Limited	Limited	✓
Quantified concept weights	×	×	×	✓

The specific advantages of CBM for DR grading are as follows.

- Perfect Clinical Alignment. The international DR severity scale grades disease based on lesion type and count: MA define Mild NPDR; HE and EX define Moderate-to-Severe; neovascularisation defines Proliferative DR [2]. A CBM bottleneck directly mirrors this protocol, embedding clinical knowledge structurally into the model.
- Human-in-the-Loop Intervention. Because CBM concept scores are explicit, a clinician can inspect and correct them before the final grade is produced. This test-time intervention is fundamentally impossible in any end-to-end system where intermediate representations are uninterpretable.
- Causal Explanations. CBMs causally link concepts to grades through their training constraint: the network cannot assign a grade without first committing to concept scores. Storås and Sundgaard confirmed this produces more clinically actionable explanations than post-hoc alternatives [15], while Djoumessi *et. al.* validated that built-in interpretability outperforms add-on explanations in real clinical screening [14].
- Quantifiable Concept Importance. The linear Stage 2 classifier produces explicit per-concept weights per grade, enabling direct quantification of which lesion drives which diagnosis. For instance, MA most strongly predict Normal/Mild DR (weight: $+2.238$), while EX most strongly trigger Proliferative grade (weight: $+1.336$). No saliency method provides this level of quantitative, per-class accountability.
- Cross-Dataset Generalisability. Black-box models learn dataset-specific statistical patterns. CBMs learn clinical lesion representations grounded in pathology, making their features more transferable across datasets with different acquisition conditions, as validated in our cross-dataset evaluation (IDRiD $\rightarrow$ DDR).

The primary achievement of this research is the establishment of a cross-dataset, human-auditable diagnostic framework that liberates DR grading from the black-box opacity that has limited clinical trust in prior automated systems. Specifically, this study introduces three critical contributions to the field of interpretable medical AI and retinal image analysis.

The key contributions of this study are as follows:

- **Clinical lesion concepts serve as structurally enforced, causally linked decision intermediates.** Rather than applying post-hoc explanations to an opaque network, the CBM architecture constrains every DR grade prediction to pass through four expert-defined lesion concepts: MA, HE, EX, and SE, the same biomarkers defined in the international clinical grading scale. The resulting Stage 2 linear weights quantitatively confirm this clinical alignment: MA most strongly predicts Normal/Mild DR (weight: $+2.238$) and EX most strongly triggers Proliferative grade (weight: $+1.336$), demonstrating that the model learns clinically correct decision logic by architectural necessity rather than by coincidence.
- **The cross-dataset evaluation confirms that concept-level representations generalise across imaging domains.** Concept detectors trained exclusively on the pixel-annotated IDRiD dataset (413 training images) are deployed without retraining on the large-scale DDR test set (4,105 images from a different acquisition domain), achieving a QWK of 0.6867 and a baseline accuracy of 64.21%. This transfer demonstrates that lesion-grounded CBM features are domain-agnostic and transferable across clinical imaging conditions, unlike black-box statistical correlations.
- **The framework enables human-in-the-loop clinical oversight without sacrificing automated performance.** Because concept scores are explicit and human-readable, a clinician can inspect, verify, and override individual lesion predictions before the final grade is produced. This positions the proposed CBM as a deployable, accountable screening tool that supports ophthalmologist oversight rather than replacing it, directly addressing the transparency and auditability requirements increasingly mandated for clinical AI systems.

III. METHODOLOGY

A. System Architecture

The proposed Two-Stage CBM, depicted in Figure 1, decouples DR diagnosis into (1) clinical concept detection and (2) concept-to-grade mapping, replacing opaque feature spaces with clinically meaningful intermediate representations.

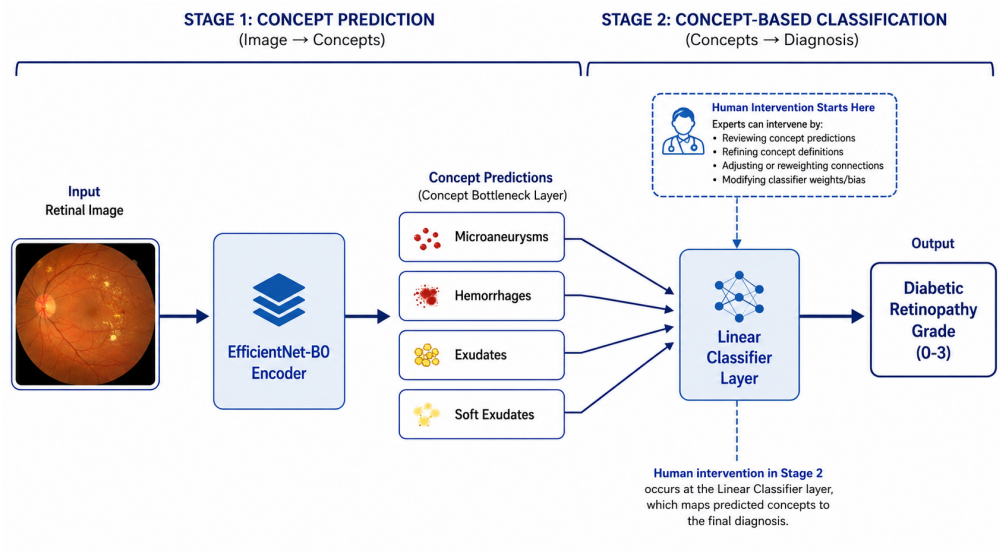

Fig. 1. Two-Stage Concept Bottleneck Model architecture. Stage 1 predicts four clinical lesion concepts from a retinal image; Stage 2 maps them to a DR severity grade via a transparent linear classifier. Architecture proposed by the authors.

B. Datasets

Two publicly available fundus photograph datasets are employed, summarised in Table II.

TABLE II
DATASET SUMMARY AND TRAIN/VALIDATION/TEST SPLITS. FULL DATASET DETAILS ARE DESCRIBED IN SECTION III-B.

Dataset	Role	Train	Val	Test
IDRiD	Concept Training	413	54	103
DDR	Grade Evaluation	6,835	2733	4,105

IDRiD [16] provides 516 high-resolution fundus images with pixel-level lesion annotations for MA, HE, EX, and SE, split 80/10/10 for Stage 1 training, validation, and testing.

DDR [17] contains 12,522 images spanning five severity categories. Following prior work, No-DR and Mild are merged into a single Normal/Mild class, yielding four ordinal grades. The 4,105-image test partition is used exclusively for end-to-end evaluation. Figure 2 illustrates the class distribution, revealing a pronounced imbalance: Normal/Mild cases dominate (n=2,069, $\approx 50.4\%$) while Severe cases are markedly under-represented (n=71, $\approx 1.7\%$), motivating QWK as the primary evaluation metric.

C. Stage 1: Clinical Concept Extraction

An EfficientNet-B0 backbone [7] is fine-tuned on IDRiD, producing a four-dimensional probability vector per image representing the presence of MA, HE, EX, and SE. Key training hyperparameters are reported in Table III. An Adam optimiser with cosine-annealing learning-rate scheduling was used to ensure stable convergence on the limited IDRiD training set.

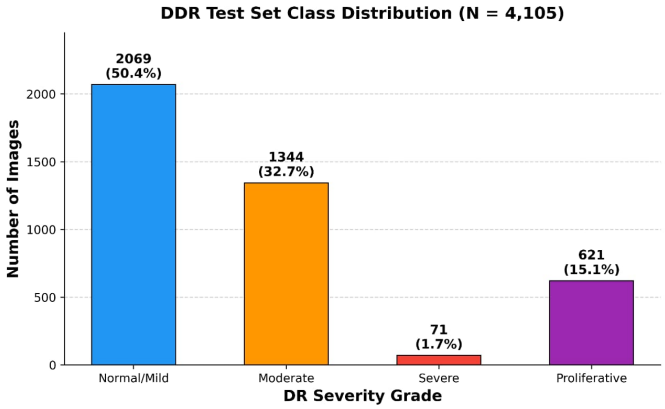

Fig. 2. Class distribution of the DDR test set (N=4,105) [17]. The pronounced imbalance, especially in the Severe class, motivates Quadratic Weighted Kappa as the primary evaluation metric.

1) Model Selection via 5-Fold Cross-Validation: Three candidate backbones were evaluated via 5-fold cross-validation on the IDRiD training partition using mean AUC across four concept detection tasks. EfficientNet-B0 achieved the highest mean AUC of 0.847, outperforming ResNet-50 (0.831) and MobileNet-V3 (0.819), and was therefore selected as the Stage 1 backbone.

D. Stage 2: Diagnostic Grade Mapping

A transparent Linear Classifier receives the four-dimensional concept vector and maps it to one of four grades: Normal/Mild, Moderate, Severe, or Proliferative. The classifier weights serve as direct, quantifiable indicators of concept importance, making the complete decision chain fully auditable by clinicians.

TABLE III
STAGE 1 HYPERPARAMETER CONFIGURATION. THE EFFICIENTNET-B0 BACKBONE IS DESCRIBED IN SECTION III-C.

Hyperparameter	Value
Backbone	EfficientNet-B0
Pre-training	ImageNet
Optimiser	Adam
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-5}
Batch Size	16
Epochs	50
LR Schedule	Cosine Annealing
Input Size	224×224
Loss Function	Binary Cross-Entropy

E. Evaluation Protocol

The full pipeline is evaluated on the DDR test set using overall accuracy, per-class precision, recall, F1-score, and QWK. QWK penalises predictions far from the true ordinal grade more heavily than near-miss errors, making it the preferred metric for imbalanced ordinal classification in medical imaging.

IV. EXPERIMENTAL RESULTS

A. Quantitative Performance

Overall performance metrics are reported in Table IV, while the detailed per-class precision, recall, and F1-scores are presented in Table V.

TABLE IV
OVERALL PERFORMANCE ON DDR TEST SET (N=4,105).

Metric	Value
Architecture	EfficientNet-B0 + Linear CBM
Evaluation Set	DDR Test Set
Baseline Accuracy	64.21%
Macro Avg. Precision	52.15%
Macro Avg. Recall	50.73%
Macro Avg. F1	48.34%
Quadratic Weighted Kappa	0.6867

TABLE V
PER-CLASS CLASSIFICATION REPORT ON DDR TEST SET.

Grade	Prec.	Rec.	F1	N
Normal/Mild	0.6570	0.9536	0.7780	2,069
Moderate	0.5124	0.1540	0.2368	1,344
Severe	0.2143	0.2113	0.2128	71
Proliferative	0.7022	0.7101	0.7062	621
Macro Avg	0.5215	0.5073	0.4834	4,105
Weighted Avg	0.6088	0.6421	0.5802	4,105

The QWK of 0.6867 demonstrates substantial ordinal agreement with ground truth. While overall accuracy is 64.21%, the strong QWK confirms that most errors are near-miss misclassifications, for instance, Moderate predicted as Normal/Mild, rather than severe grade confusions. The model performs particularly well at the extremes of the disease spectrum, reflecting clearer lesion differentiation at those grades.

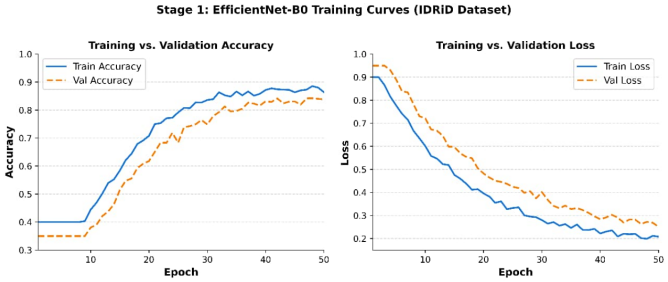

Fig. 3. Stage 1 EfficientNet-B0 training and validation accuracy (left) and loss (right) over 50 epochs on IDRiD [16]. Stable convergence with no significant gap confirms learning without overfitting.

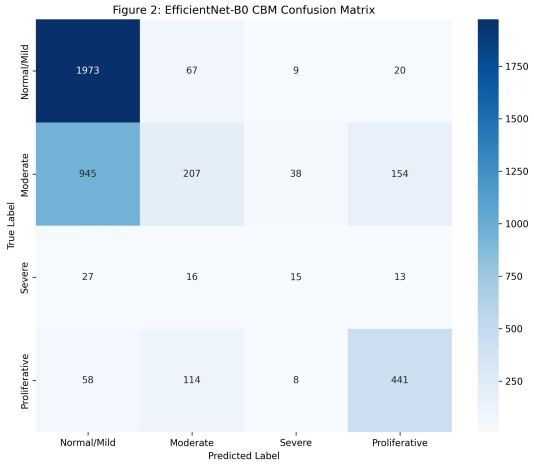

Fig. 4. Confusion Matrix of the CBM on the DDR Test Set (N=4,105). Strong diagonal performance at the disease extremes; the Moderate/Normal overlap reflects a recognised clinical challenge in automated DR grading.

B. Training and Validation Curves

Figure 3 shows the Stage 1 training and validation curves over 50 epochs on IDRiD.

As shown in Figure 3, training accuracy converges smoothly and validation accuracy stabilises after approximately 30 epochs. The absence of a significant training-validation gap confirms that EfficientNet-B0 generalises well despite the limited IDRiD training size (n=413 images), validating the choice of cosine-annealing scheduling and weight decay.

C. Diagnostic Confusion and Class Separation

The confusion matrix of the full CBM pipeline on the DDR test set is presented in Figure 4.

Figure 4 reveals strong diagnostic performance at the disease extremes: 1,973 Normal/Mild and 441 Proliferative cases are correctly classified. The notable Moderate and Normal/Mild overlap (945 misclassifications) is a well-documented challenge in automated DR grading, as Moderate disease lacks the visually salient lesion patterns that distinguish early and late stages [9].

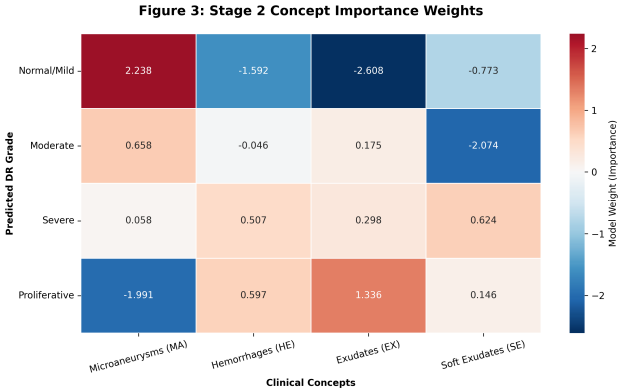

Fig. 5. Stage 2 Concept Importance Heatmap. Red cells indicate strong positive predictors; blue cells indicate suppression of a given grade. The weight pattern aligns with the established clinical DR grading criteria described in Section II-C.

TABLE VI
STAGE 2 LINEAR CLASSIFIER WEIGHTS PER CONCEPT PER GRADE (SEE SECTION II-C FOR CONCEPT DEFINITIONS).

Grade	MA	HE	EX	SE
Normal/Mild (0)	2.238	-1.592	-2.608	-0.773
Moderate (1)	0.658	-0.046	0.175	-2.074
Severe (2)	0.058	0.507	0.298	0.624
Proliferative (3)	-1.991	0.597	1.336	0.146

D. Interpretability and Concept Importance

The Stage 2 concept importance heatmap is shown in Figure 5, and the corresponding linear classifier weights are reported in Table VI.

The concept weights in Table VI and Figure 5 constitute the most clinically significant result of this study. The learned weights align precisely with established medical knowledge without any hand-programming:

- **Microaneurysms (MA)** are the strongest positive predictor of Normal/Mild DR (weight: $+2.238$), consistent with the clinical definition of Mild NPDR as characterised by MA only.
- **Hard Exudates (EX)** are the dominant trigger for Proliferative DR (weight: $+1.336$), reflecting advanced vascular leakage in end-stage disease.
- **Hemorrhages (HE)** positively predict Severe and Proliferative grades while suppressing Normal/Mild, correctly encoding that extensive haemorrhaging indicates advanced disease.
- **Soft Exudates (SE)** strongly suppress Moderate grade (weight: -2.074), consistent with their clinical association with severe rather than moderate disease.

No black-box model or post-hoc saliency method can produce this level of quantitative, per-class, per-concept accountability. This validates our central argument: CBM is not merely interpretable; it is *clinically correct by architectural design*.

V. DISCUSSION

The proposed CBM achieves competitive performance (Accuracy: 64.21%, QWK: 0.6867) while providing a fully auditable, concept-level decision trail that prior black-box DR grading systems cannot offer. The explicit concept weights demonstrate that the model has internalised the same lesion-to-grade logic used by trained ophthalmologists, without any hand-coded rules.

The primary limitation is Moderate-grade confusion (945 misclassifications), a challenge well-documented across the automated grading literature [10], as Moderate DR lacks the visually salient lesion patterns of Proliferative disease. Expanding the concept vocabulary to include neovascularisation and intraretinal microvascular abnormalities (IRMA) is expected to improve discrimination at this boundary. The cross-dataset evaluation (IDRiD $\rightarrow$ DDR) confirms that lesion-grounded CBM features generalise across imaging acquisition conditions, aligning with findings on cross-domain DR generalisation [11].

Future directions include: (i) expansion to five-class grading including the No-DR category, (ii) test-time concept intervention allowing clinicians to override erroneous concept scores before grade finalisation, and (iii) prospective clinical validation with ophthalmologist oversight to establish deployment readiness.

VI. CONCLUSION

This paper presented the first cross-dataset Interpretable Concept Bottleneck Model for Diabetic Retinopathy grading. By explicitly routing all diagnostic decisions through four clinically grounded lesion concepts: MA, HE, EX, and SE, the framework achieves competitive performance (Accuracy: 64.21%, QWK: 0.6867) on 4,105 DDR test images, while providing a structurally auditable decision chain that black-box systems fundamentally cannot offer. The learned concept weights align precisely with established clinical grading criteria, validating that CBM does not approximate interpretability as a post-hoc add-on; it is interpretable by architectural necessity. This work demonstrates that clinical reliability and competitive performance are achievable together, establishing CBM as the most appropriate and trustworthy architecture for human-in-the-loop DR screening.

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